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Computer Networks Mini Project

Report: Network Design

Tribhuvan International Airport (TIA), Kathmandu

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1 Introduction

This report describes the mini-project work carried out for the Computer Networks course: the design and realisation, in Cisco Packet Tracer, of a complete enterprise network for Tribhuvan International Airport (TIA), the only international airport of Nepal. A single airport site must carry several categories of traffic that differ sharply in scale and sensitivity - passenger Wi-Fi, check-in and boarding, airline offices, surveillance cameras, baggage handling, immigration and customs, cargo, and administration - and all of them must share one address block while remaining correctly routed and, where required, separated.

Separation inside the terminal is achieved with VLANs, inter-VLAN routing is provided by router sub-interfaces, and the zones are joined by a multi-area OSPF backbone that reaches the Internet through an edge router towards a single upstream service provider. Every internal router is named with the author's first name and an index, as the submission brief requires.

1.1 Objectives

1. Divide a single /20 address block among ten networks of at least six different sizes using variable length subnet masking (VLSM).
2. Build a routed network of ten routers, of which three carry no user LAN, running OSPF in four areas including area 0.
3. Provide a default route towards the upstream provider and arrange for return traffic without any dynamic routing on the provider side, using the minimum number of route entries.
4. Provide DHCP on every LAN, two DNS servers in different LANs, a further DNS server inside the provider network, and two web servers on different subnets.
5. Realise the whole design in Packet Tracer with working configurations, remote access by telnet with the required passwords, and a fully labelled topology.

2 Design Overview

The network is arranged in three tiers. Ayushma1 is the edge router; it connects to the provider over a serial leased line and originates the default route. Ayushma2, Ayushma3 and Ayushma4 form the OSPF area 0 backbone. Five distribution routers hang below the backbone, each owning its own LANs (and, in the terminal, VLANs), its own DHCP pools and its own OSPF area; the Terminal and Security areas each reach the backbone over two independent paths so that no single link failure isolates them. The completed network contains 10 routers (9 internal, 1 provider), 11 switches, 7 servers and representative PCs in every LAN.

Router	Role	Function	OSPF area
Ayushma1	Edge	WAN uplink to the ISP; originates the default route; no user LAN	0
Ayushma2	Core	Backbone core; hosts Server Farm A	0
Ayushma3	Core	Backbone transit; no user LAN	0
Ayushma4	Transit	Backbone transit towards Areas 2 and 3; no user LAN	0
Ayushma5	Terminal	Area Border Router; hosts Server Farm B	0 & 1
Ayushma6	Terminal	Inter-VLAN routing for VLAN 10, 20, 30 (router-on-a-stick)	1
Ayushma7	Security	Area Border Router; CCTV LAN	0 & 2
Ayushma8	Operations	Immigration and Baggage LANs	2
Ayushma9	Cargo	Area Border Router; Cargo and CAAN Admin LANs	0 & 3
Ayushma10	Provider	ISP router; top-level DNS; single static return route	-

Table 1: Router naming and roles. Ayushma1, Ayushma3 and Ayushma4 carry no user LAN.

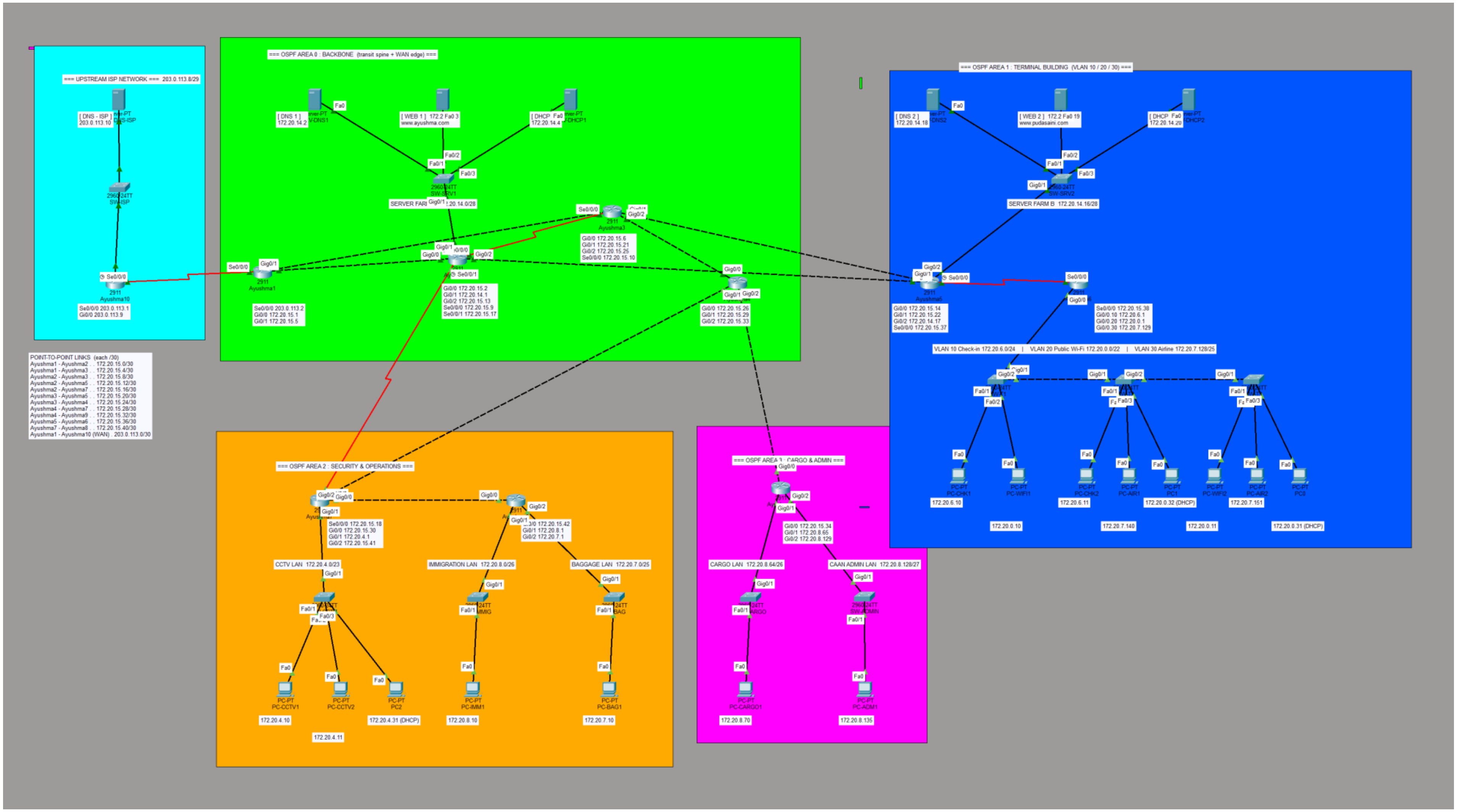

Figure 1: Labelled topology as realised in Packet Tracer, colour-coded by OSPF area. Labels give the interface addresses of every router, the network ID of every LAN and VLAN, the IP address of every PC, the service and address of every server, and the network ID of all twelve point-to-point links.

3 Addressing

The airport is addressed from a single private block, 172.20.0.0/20 (4,096 addresses). Subnets were allocated with VLSM in descending order of size so that no space is wasted, using seven different prefix sizes. The ten LANs are given below.

Zone (VLAN)	Network ID	Usable range	Hosts	Router
Public Wi-Fi (VLAN 20)	172.20.0.0/22	.0.1 - .3.254	1022	Ayushma6
CCTV	172.20.4.0/23	.4.1 - .5.254	510	Ayushma7
Check-in (VLAN 10)	172.20.6.0/24	.6.1 - .6.254	254	Ayushma6
Baggage	172.20.7.0/25	.7.1 - .7.126	126	Ayushma8
Airline (VLAN 30)	172.20.7.128/25	.7.129 - .7.254	126	Ayushma6
Immigration	172.20.8.0/26	.8.1 - .8.62	62	Ayushma8
Cargo	172.20.8.64/26	.8.65 - .8.126	62	Ayushma9
CAAN Admin	172.20.8.128/27	.8.129 - .8.158	30	Ayushma9
Server Farm A	172.20.14.0/28	.14.1 - .14.14	14	Ayushma2
Server Farm B	172.20.14.16/28	.14.17 - .14.30	14	Ayushma5

Table 2: The ten LANs, using prefix sizes /22, /23, /24, /25, /26, /27 and /28.

The remaining space carries eleven router-to-router links as /30 subnets numbered from 172.20.15.0. A /30 is used for a link between two routers because it provides exactly two usable addresses and wastes nothing. The WAN link to the provider uses 203.0.113.0/30 and the provider LAN uses 203.0.113.8/29. Overall utilisation is roughly 2,220 of 4,096 addresses, leaving about 46% free for growth.

4 Routing Configuration

4.1 Interior routing

OSPF process 1 runs across four areas. Area 0 is the backbone and every other area touches it directly, as the protocol requires. Ayushma5, Ayushma7 and Ayushma9 are Area Border Routers and summarise routes between their area and the backbone. Router IDs are pinned manually (1.1.1.1 to 9.9.9.9) so that a router keeps a stable identity even if a physical interface is renumbered or fails.

4.2 Exterior routing

Ayushma1 forwards all Internet traffic to the provider using a single default route, which is advertised into OSPF so that no other router in the airport needs a static route to reach the Internet:

```
ip route 0.0.0.0 0.0.0.0 203.0.113.1
router ospf 1
 default-information originate
```

4.3 Return path without dynamic routing

The provider runs no routing protocol with the airport. Because the whole airport occupies one contiguous /20 block, the provider needs exactly one route entry to return all traffic - the minimum possible:

```
ip route 172.20.0.0 255.255.240.0 203.0.113.2
```

No address translation is used: the provider holds a route to the airport's block, so internal hosts reach the provider network directly. This is the simplest arrangement that satisfies the brief's requirement to return traffic without dynamic routing using the fewest route entries.

5 Services

DHCP. Each distribution router provides DHCP for the LANs it serves, holding one pool per LAN that supplies the IP address, subnet mask, default gateway and DNS server, with the gateway and the static server addresses excluded from each pool. Two DHCP servers are also placed in the server farms. This was tested by connecting a PC set to obtain an address automatically, which received 172.20.4.31 on the CCTV LAN with the correct gateway and DNS server.

DNS and Web. Two internal resolvers, SRV-DNS1 and SRV-DNS2, sit in different LANs and hold identical A records, so either can resolve the airport's own web names; a third resolver in the provider network represents external DNS. Two web servers on different subnets host www.ayushma.com and www.pudasaini.com. Because the resolvers are internal, clients reach the web servers directly rather than sending traffic out to the edge and back.

Server	Address	Service	Notes
SRV-WEB1	172.20.14.3	HTTP	www.ayushma.com, Server Farm A
SRV-DNS1	172.20.14.2	DNS	Primary resolver, Server Farm A
SRV-DHCP1	172.20.14.4	DHCP	Server Farm A
SRV-WEB2	172.20.14.19	HTTP	www.pudasaini.com, Server Farm B
SRV-DNS2	172.20.14.18	DNS	Second resolver in a different LAN
SRV-DHCP2	172.20.14.20	DHCP	Server Farm B
SRV-DNS-ISP	203.0.113.10	DNS	Resolver inside the provider network

Table 3: Servers and the services they provide.

6 Device Access Configuration

Every router - and every switch - is configured for remote access by telnet with the passwords required by the brief. Each switch was also given a management IP address in its LAN so that it is reachable:

```
enable secret class
line con 0  /  password cisco  /  login
line vty 0 4  /  password network  /  login  /  transport input telnet
```

The console password is cisco, the privileged (enable) password is class, and the telnet password is network. An identifying banner was also configured. Telnet was tested by opening a session from a PC to a router (telnet 172.20.6.1), entering network at the login prompt and class to reach privileged mode.

7 Testing and Verification

Check	Result
OSPF	Adjacencies reach the FULL state across all four areas
Interface state	All routed interfaces and trunks up/up; health check reports no down links
Reachability	Every internal network answered pings from PCs in different areas (Area 1 and Area 3)
Internet forwarding	Traffic to an outside address is correctly forwarded to the provider via the default route
DHCP	A client received a lease with the correct gateway and DNS from its LAN's pool (172.20.4.31)
DNS / Web	Records added on both resolvers; www.ayushma.com and www.pudasaini.com resolve; HTTP enabled
Telnet	telnet with network then class succeeds on every router
Provider routing	No routing protocol on the provider; it holds exactly one static route entry
Addressing	No subnet overlap and no duplicate IP addresses; every /30 correctly paired
Redundancy	Shutting the primary link of Areas 1 and 2 caused OSPF to re-route through the backup automatically

Table 4: Results measured on the built topology.

8 Discussion

The shape of the network came mostly from two requirements pulling in the same direction: having some routers with no user LAN, and using more than one OSPF area. Together these meant the design could not be flat, so it ended up as a backbone with the different airport zones hanging off it. The mix of serial and Ethernet links also came from a practical limit - a 2911 router has only three Gigabit ports, so once a core router used all three, its extra links had to be serial.

The main problem I ran into was while testing DHCP. When I added a new PC to one of the terminal switches it did not get an address and ended up with an APIPA (169.254) address instead. On checking, the reason was that the switch port was still in the default VLAN 1, which has no gateway and no DHCP pool, so the request had nowhere to go. Putting the port into the correct VLAN fixed it straight away, and it made clear that on a switch carrying VLANs the port has to be in the right VLAN before a device can get an address at all.

9 Conclusion

The complete network for Tribhuvan International Airport was designed and built in Cisco Packet Tracer, and all the requirements of the assignment were met. The 172.20.0.0/20 block was divided using VLSM into ten LANs of seven different sizes; ten routers run OSPF across four areas, with three of them carrying no user LAN. The Internet is reached through a single default route to the provider, and the provider returns traffic using one static route with no dynamic routing. DHCP, DNS and web services were configured and found working, every router and switch is reachable by telnet with the required passwords, and the whole topology is labelled and colour-coded by area.

Testing with pings from computers in different areas confirmed that every internal network is reachable from any computer. Working through the design also made a few ideas much clearer in practice - in particular how the VLAN of a switch port decides whether a device can get an address from DHCP, and how a single default route together with one static return route keeps the internal and external routing cleanly separated. Overall the project met all of its objectives.